DELIN QU

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Physical AI ◇ Video Action Model ◇ Physical Reasoning Models

My current research focuses on system design and emergent capabilities in agentic physical AI.
I build robots that reason about the physical world to steer fast actions and adapt in context.

EDUCATION

Fudan University — *Sep. 2022 – Jun. 2027*
Advisor: Prof. Xuelong Li
Ph.D. in Computer Science and Technology — Shanghai AI Laboratory

Hunan University — *Sep. 2018 – Jun. 2022*
B.Eng. in Computer Science and Technology — GPA: 3.86/4.00; rank: 1/208

SELECTED PUBLICATIONS

- NVIDIA. **Cosmos 3: Omnimodal World Models for Physical AI**. *arXiv preprint, 2026; contributor.*
- Shalfun Li*, *et al.*, **Delin Qu***, ... Qian Wang. **WALL-WM: Carving World Action Modeling at the Event Joints**. *arXiv preprint, 2026; core contributor.*
- **Delin Qu***, Haoming Song, Qizhi Chen, *et al.*, Bin Zhao, Dong Wang, Xuelong Li. **EO-1: Interleaved Vision-Text-Action Pretraining for General Robot Control**. *IEEE TPAMI, 2026.*
- **Delin Qu***, Haoming Song, Qizhi Chen, *et al.*, Dong Wang, Xuelong Li. **SpatialVLA: Exploring Spatial Representations for Visual-Language-Action Models**. *RSS, 2025, Spotlight; 524 citations.*
- **Delin Qu***, Jiaru Li, Bin Zhao, Dong Wang, Xuelong Li. **PiX: Pixel Space World Action Model**. *TPAMI, under review.*
- **Delin Qu***, Chi Yan, Dong Wang, *et al.*, Bin Zhao, Xuelong Li. **Implicit Event-RGBD Neural SLAM**. *CVPR, 2024, Spotlight (top 2.6%).*
- **Delin Qu***, Qizhi Chen, *et al.*, Zhigang Wang, Dong Wang, Xuelong Li. **LiveScene: Language Embedding Interactive Radiance Fields for Physical Scene Rendering and Control**. *NeurIPS, 2024.*
- **Delin Qu***, Bangyan Liao*, Huiqing Zhang, Omar Ait-Aider, Yizhen Lao. **Fast Rolling Shutter Correction in the Wild**. *TPAMI, 2023.*
- **Delin Qu***, Yizhen Lao, *et al.*, Dong Wang, Bin Zhao, Xuelong Li. **Towards Nonlinear-Motion-Aware and Occlusion-Robust Rolling Shutter Correction**. *ICCV, 2023.*
- Bangyan Liao*, **Delin Qu***, Yifei Xue, Huiqing Zhang, Yizhen Lao. **Revisiting Rolling Shutter Bundle Adjustment: Toward Accurate and Fast Solution**. *CVPR, 2023.*
- Haoming Song*, **Delin Qu***†, *et al.*, Bin Zhao, Dong Wang, Xuelong Li. **Hume: Introducing System-2 Thinking in Visual-Language-Action Models**. *CVPR, 2026;* † Project Lead.
- Qizhi Chen*, **Delin Qu***†, *et al.*, Dong Wang, Bin Zhao, Xuelong Li. **FreeGaussian: Guidance-Free Controllable 3D Gaussian Splats with Flow Derivatives**. *AAAI, 2026, Oral (top 4.6%);* † Project Lead.
- Chi Yan*, **Delin Qu***†, *et al.*, Zhigang Wang, Dong Wang, Xuelong Li. **GS-SLAM: Dense Visual SLAM with 3D Gaussian Splatting**. *CVPR, 2024, Spotlight (top 2.6%); 710 citations;* † Project Lead.

WORK EXPERIENCE

Research Intern, Cosmos 3 & Cosmos 3.5 — Jun. 2026 – Present
NVIDIA · *Advised by Ming-Yu Liu* — *Santa Clara, USA*
- Build unified action spaces for cross-embodiment robot-policy mid-training in Cosmos 3 and Cosmos 3.5.
- Design in-context learning capabilities for generalist robot policies.

Research Intern, WALL-WM — Sep. 2025 – Dec. 2025
X-Square Robot · *Advised by Xiaofan Li* — *Shenzhen, China*
- Core contributor to WALL-WM; built pre-training, fine-tuning, and alignment pipelines for world-action models, with real-robot evaluation focused on generalization and deployment robustness.

Research Intern, OpenSS2 — Apr. 2025 – Jun. 2025
AgiBot · *Advised by Guanghui Ren* — *Shanghai, China*
- Proposed OpenSS2, a hierarchical System-1/System-2 policy using phased flow matching and value-guided test-time selection, improving task success by 24% over π_0 on AgiBot G1.

Joint Ph.D. Researcher, EO-Robotics / IPEC Aug. 2022 – Present
Shanghai AI Laboratory · *Advised by Xuelong Li* *Shanghai, China*

- Lead end-to-end research on generalist embodied foundation models, from spatially grounded VLA pre-training (SpatialVLA) and interleaved vision-text-action learning (EO-1) to System-2 control (Hume).
- Collaborated with Hugging Face's LeRobot team to integrate EO-1 into the official LeRobot codebase.
- Drive multimodal data curation, distributed pre-/post-training, and simulation-to-real robot evaluation.
- Mentor junior researchers and interns in project design, implementation, experimentation, and paper writing.

Teaching Assistant 2022 – 2026
Fudan University *Shanghai, China*

- COMP130135: Object-Oriented Programming
- COMP620002: Artificial Intelligence

SELECTED PROJECTS

OpenPI Comet *Core Contributor*
Led development of Team Comet's end-to-end $\pi_{0.5}$ training system across multi-dataset pre-training, rejection-sampling post-training, data generation, and evaluation; placed 2nd worldwide in the 2025 BEHAVIOR Challenge, only 0.0085 Q-score behind the winning team.

FastUMI *Core Contributor*
Co-developed a scalable, hardware-independent Universal Manipulation Interface for rapid deployment, robust real-world robot data acquisition, and reusable processing workflows.

Any4LeRobot *Core Contributor* ◇ *1.1k stars*
Developed data-conversion, preprocessing, version-migration, and multi-dataset training utilities spanning Open X-Embodiment, AgiBot, RoboMIND, RoboCasa, LIBERO, and RLDS for the LeRobot ecosystem.

BOOK

Bin Zhao, Dong Wang, **Delin Qu**, Haoming Song, Xuelong Li. *Embodied Intelligence: Technology Frontiers and Applications.* Chinese-language book manuscript; under publisher review.

FUNDING & AWARDS

National Natural Science Foundation of China (NSFC) 2024
Learning to Grow: Interactive Developmental Morphology for Embodied Manipulation Policy

- **Question:** What growth-oriented learning and multi-source interaction mechanisms enable embodied manipulation policies to continually improve in real-world environments?
- **Approach:** Develop a spatial-prior-guided action model that continually improves through a closed loop of planning, execution, feedback, and self-exploration.

Tencent Scholarship (top 1‰) 2023
National Scholarship (top 1‰) 2020, 2021
2025 BEHAVIOR Challenge, 2nd Place Worldwide 2025

INVITED TALKS

Real-Time Interactive World Action Models Jun. 2026
Shanghai AI Laboratory *Shanghai, China*
Exploring Spatial Representations for Vision-Language-Action Models Mar. 2025
Institute of Artificial Intelligence, TeleAI / China Telecom *China*

PROFESSIONAL SKILLS

Peer review: CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, TPAMI, AAAI; AAAI Program Committee member.

Physical AI models: VLA models, world models, multimodal pre-training and post-training, in-context learning

Learning systems: Python, PyTorch, Hugging Face Transformers, distributed training, data curation and evaluation

Robot learning: Cross-embodiment action spaces, flow matching, diffusion policies, action chunking, test-time computation

3D perception: Gaussian Splatting, neural rendering, visual SLAM, real-robot evaluation